Parallel and HPC
HW4 Report

Link to github repository:

https://github.com/DMelkonyan111/Parallel-and-HPC-Homeworks

I implemented four versions of the DNA counting problem: scalar, multithreading, SIMD, and SIMD with multithreading. One character at a time is processed by the scalar version. I split the sequence into four sections for the multithreading version, with a thread handling each segment. In the combined version, I used SIMD within each thread, while the SIMD version compares 32 characters simultaneously using AVX2.I changed lowercase letters to uppercase in order to solve the character buffer issue. SIMD processes several characters simultaneously, the multithreading version divides the buffer among threads, and the scalar version scans characters sequentially. SIMD is used in each thread of the combined version. I handled the remaining characters in a sequential manner and distributed buffers equally among threads. SIMD handles several characters at once, which speeds up processing. According to my performance measurements, multithreading provides a moderate speedup, SIMD alone further enhances it, and SIMD plus multithreading yields the best speed, particularly for large buffers.

```
davitm@ubuntu-aua-os:~/workshop/Homeworks/4$ ./EX1
DNA size: 100 MB
Threads used: 4


Counts (A C G T):
0 0 0 0

Scalar time:                 1.093 sec
Multithreading time:         1.105 sec
SIMD time:                   0.065 sec
SIMD + Multithreading time: 0.057 sec
```

```
davitm@ubuntu-aua-os:~/workshop/Homeworks/4$ ./EX2
Buffer size: 100 MB
Threads used: 4

Multithreading time:      0.546 sec
SIMD time:                0.556 sec
SIMD + Multithreading:    0.495 sec
```